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Maternal Discrimination and the Development of Sex Differences in Exploratory Behaviour in Infant Spiny Mice (*Acomys cahirinus*)

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The present paper reports sex differences in exploratory behaviour by infant Spiny Mice, *Acomys cahirinus*, that may, in part, be related to differences in maternal behaviour towards pups; like some other rodents, mother *Acomys* differentiate behaviourally between male and female pups. In Experiment 1 infant *Acomys* were allowed to explore a novel arena. This experiment showed that even by Day 3 (the day of birth = Day 1) female *Acomys* explored a novel environment more than males; they entered the arena sooner than males and spent more time in contact with a novel object. Experiment 2 showed that infant females were more active than males when observed in the home cage in the presence of their parents and made more approaches to the mother. Mothers, on the other hand, directed more licking behaviour towards males. Experiment 3 focused on the exploratory behaviour of individual pups in the presence of the mother. Given access to a large, complex arena, female pups explored more than males. The results also showed that mothers direct more of their social interactions towards sons than daughters, particularly when pups are about a week old. Some mothers appear to "direct" the movement of their offspring, by blocking their forward movement; this was done more often to male than to female pups. The data suggest that the previously observed changes in exploratory behaviour at this time, and the emergence of sex differences in exploration, may in part depend upon the mothers' reactions to pups by sex.

Exploratory behaviour is usually defined, and studied, in relation to the physical environment, typically using measures of locomotor activity or responses to novel objects (see review by Birke & Archer, 1982). But for any

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social animal, the environment includes conspecifics: for young mammals in particular, exploration may be expected to occur in relation to the mother, alongside her, for instance, or by making short forays away from her. Any description of developmental changes in exploratory behaviour will, therefore, need to take into account social changes—in particular, in the relationship between infant and mother.

The present paper reports an investigation of the relationship between exploratory behaviour and social behaviour during early development in infant Spiny Mice (*Acomys cahirinus*). *Acomys* are precocial; the young are born with eyes open and capable of moving around; because of their precocity, young spiny mice are likely to explore their environment actively from a very young age.

Because adult male and female *Acomys* tend to differ in the general pattern of movements used when exploring (Birke & Sadler, 1986), we have begun to investigate possible sex differences in exploratory behaviour of this species during infancy. Confronted with an unfamiliar environment (in a pilot experiment), infant females tended to explore more than males, the sex difference appearing when animals were first tested at 3 days old.

Behavioural sex differences appearing very early in infancy may be correlated directly with endogenous hormone levels. However, parental behaviour towards pups of different sex could also be a contributory factor. Rat mothers, for example, discriminate between male and female pups on the basis of olfactory cues that may in turn depend upon differences in hormone levels (Moore, 1982; Birke & Sadler, 1987a). Specifically, male rat pups are licked significantly more (similar findings have been reported for mouse pups, *Mus musculus*: Alleva, Caprioli, & Laviola, 1989).

Such findings raise the possibility that any sex differences occurring among infants while they are young enough still to suckle may be mediated, at least partly, through differences in the mother's behaviour towards pups of each sex. The main study reported here set out to investigate the relationship between maternal behaviour towards the pups and the pups' developing exploratory behaviour; our initial hypothesis was that differences in the way that male and female infants explore a new environment would be correlated with differences in the behaviour of mothers towards pups of each sex. Before embarking on the main study, however, we conducted two preliminary experiments. The first involved a replication of the pilot study, investigating the extent of sex differences in exploratory behaviour of infants tested alone. In the second experiment, we conducted experimental observations of social interactions between relatively undisturbed animals in their home cages; the purpose of this experiment was to establish whether maternal (or paternal) discrimination occurs in familiar surroundings, analogous to the situation used in studies of rat mothers, before testing animals (with mothers) in unfamiliar surroundings.

EXPERIMENT 1

Method

Subjects. Twenty-four infant *Acomys* were used for this experiment, half of each sex, from 16 litters. Where litters were born consisting of three or more pups, only one or two from each litter were used in the experiment, to reduce litter effects. All animals were housed in cages 25 × 40 cm, with pine shavings as bedding. They were fed rat chow and sunflower seeds, with water ad libitum, and a twice-weekly supplement of pieces of apple. All animals were bred in the Animal House of the Open University and were maintained on a reversed lights schedule (lights on from 2100 to 0900 hr).

Apparatus. Animals were placed into a start-box (15 × 5 × 5 cm) which led into a small arena (40 × 40 × 15 cm), made of clear Perspex with a similar lid. In the centre of the arena was a novel object, a tube 15 cm high and 5 cm in diameter, placed vertically. In order to limit the animals' access to the centre, the object was placed inside a square box made of clear Perspex, 12 × 12 × 15 cm; in the centre of each face was a hole 5 cm in diameter, 3 cm from the floor, through which the animal could enter the centre.

Procedure. Behavioural tests took place between 0915 and 1500 hr, when the animals were most active, over the same period of testing as that used in our previous experiments (testing on Days 3, 7, 10, 14, 17, and 21 after birth; the day of birth = Day 1). On Day 2, all animals were handled and weighed (for routine data collection) and then introduced into the test apparatus for 2 min each, to reduce the effect of complete novelty on the first test.

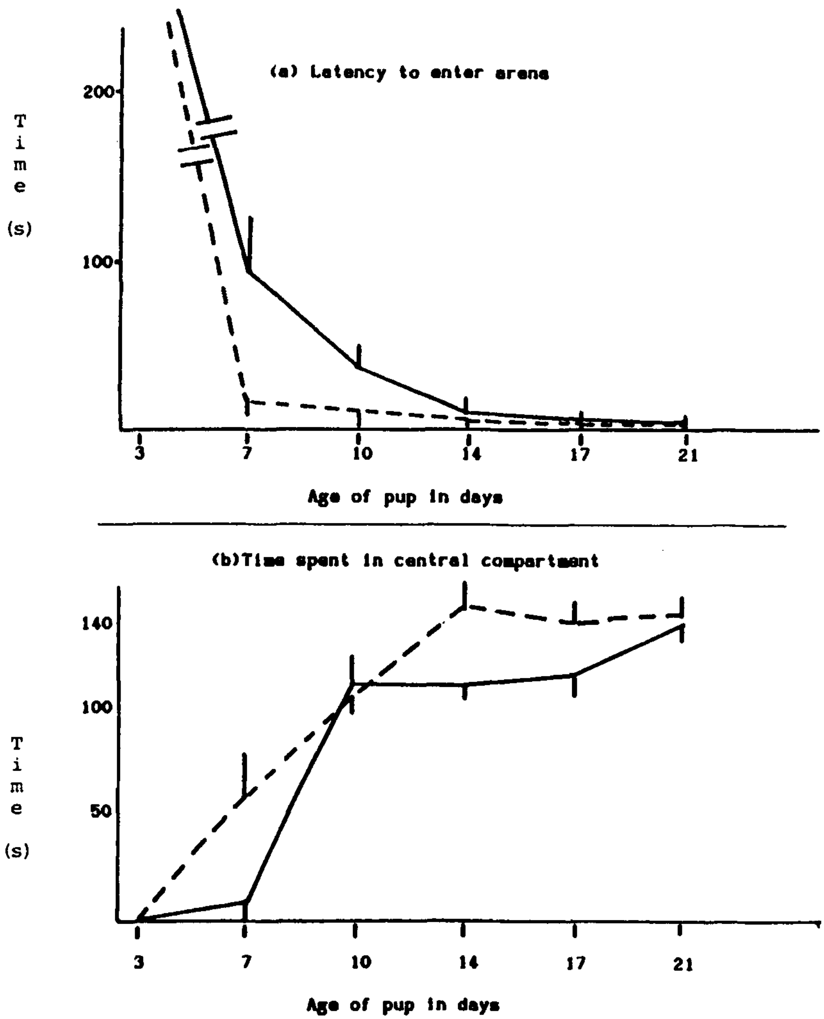
In all tests, the animal remained in the start-box with the door into the arena closed, for 2 min. The door was then removed, allowing the animal access into the arena. Behaviour was then recorded using an Apple II microcomputer to measure the following categories:

- latency to leave the start-box; frequency of re-entries to the start-box;
- latency to enter the central compartment; and duration of time spent in the centre (these two measures are not independent);
- frequency of entries to the central square and contacts with the object;
- number of times the animal was observed in the corners of the arena, and numbers of times it was observed making contact with the corners of the inner box. These last two measures were included because previous work had shown that adult females tend more than males to adopt a strategy of exploration consisting of rapid dashes around the periphery of a large arena (Birke & Sadler, 1986).

At the end of 10 min, recording stopped, and the animal was returned to its homecage. The arena was then washed down with detergent and air-dried. If another animal from the same litter was tested on the same day, at least 15 min was left between animals.

Results

Data are summarized in Figure 1 (a–d) and (e–h). For analysis, all latency/ duration measures were normalized by log-transformation; data were then



analysed using two-way analyses of variance, with repeated measures for the age variable. Changes with age were significant at the 0.01 level for all variables measured, $F(5, 120)$ varied between 22.9 and 75.7. Four measures showed significant effects of sex: females left the start-box significantly sooner than males, $F(1, 22) = 5.24$, $p < 0.05$. Females also spent longer in the

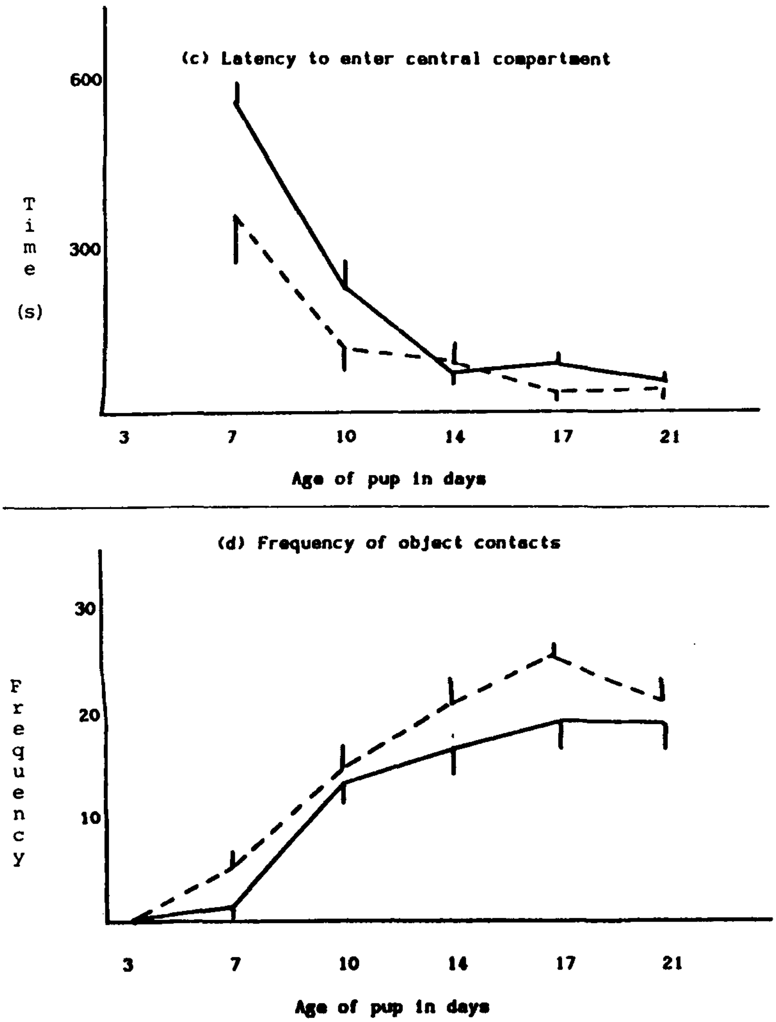
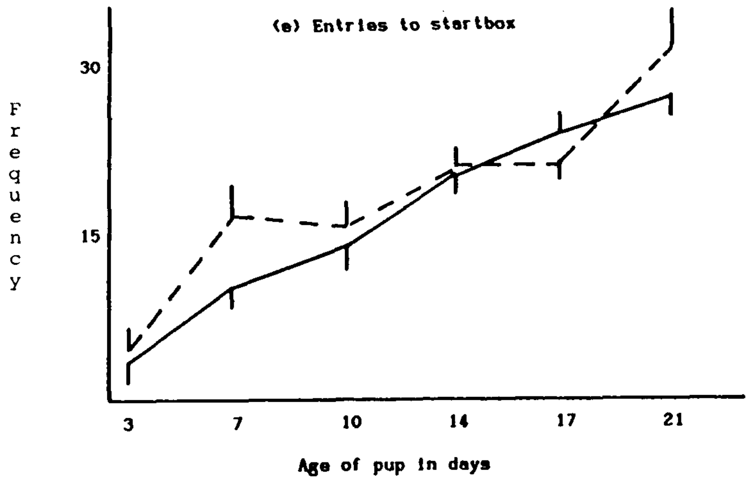
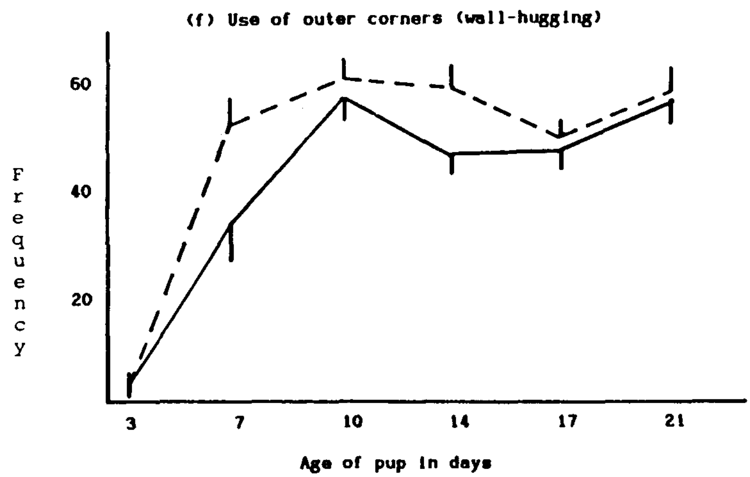


FIG. 1 (a–d). Exploratory behaviour by infant Spiny Mice: latencies to enter compartments and behaviour in central compartment (time spent there and frequency of object contact). Solid lines: males ($n = 12$); dotted lines: females ($n = 12$). Mean frequency $\pm$ s.e.m.





central box, $F(1, 22) = 4.9$, $p < 0.05$, and made contact with the central object sooner, $F(1, 22) = 9.8$, $p < 0.01$. Females also contacted the central object more frequently, $F(1, 22) = 4.8$, $p < 0.05$. There were no significant interactions.

Discussion

The data suggest that there are sex differences in the responses of infant *Acomys* to novel stimuli, as the pilot study had indicated. Young males (Days 3 and 7) were more hesitant to enter the arena and accordingly spent less time

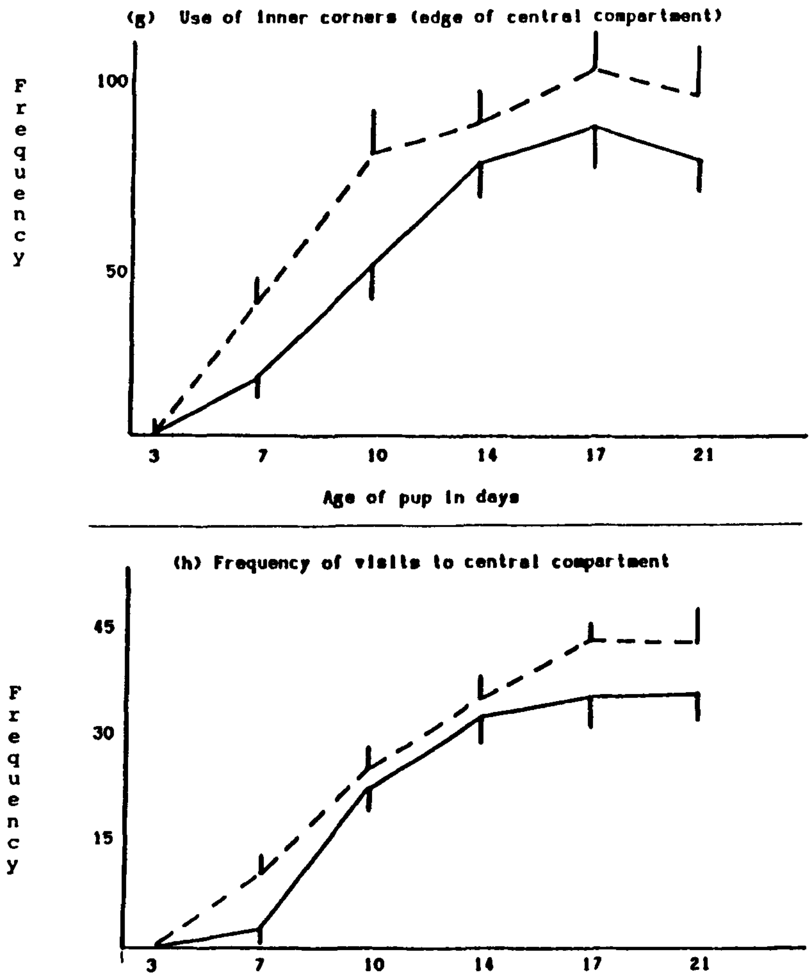


FIG. 1 (e–h). Frequency of contact with different parts of apparatus during exploration. Solid lines: males ($n = 12$); dotted lines: females ($n = 12$). Mean frequency ± s.e.m.

in (and paid fewer visits to) the central compartment. Between days 7 and 14, as the animals became more mobile, there was a rapid increase in behaviour that might be described as exploratory, namely contacts with objects and movements around the central box, and a concomitant decrease in latency to contact the object.

Different rates of maturation could be one explanation for the sex difference in response to novelty, although that seems unlikely. We have found no evidence of sex differences in the rate of weight gain (unpublished observations), and all *Acomys* pups are born relatively well advanced (with eyes open, for example). In the next two experiments, we addressed another possibility, that the different willingness of males and females to explore may reflect differences in social experience, particularly interactions with the mother. Experiment 2 was designed to investigate possible sex differences in behaviour in the familiar surroundings of the home cage.

EXPERIMENT 2

Methods

Subjects. Thirty-two infants from twenty litters of *Acomys* were used in this experiment, 17 males and 15 females, housed as for Experiment 1.

Procedure. In order to focus upon the period during which previous changes were most marked, animals were tested on Days 2, 4, 6, 8, and 10. Prior to any observation, a note was made of where in the cage the mother tended to nurse her young: although *Acomys* does not make a nest, we have observed that they do tend to prefer particular places in the cage for nursing, usually underneath the food hopper. One or two pellets of food were placed on the floor of the cage, to enable pups to feed independently. For testing, the home cage was removed from the rack and placed on the floor vertically below a videocamera. All pups were then removed and placed in a holding cage in the furthest corner of the room. After 2 min, one pup was replaced quietly in the site previously designated as the nursing area. As male Spiny Mice do show parental behaviour toward pups (Makin & Porter, 1984), both parents were left in the cage.

Observations then began, using a Panasonic videocassette recorder and a camera sensitive to low light intensities. All observations were conducted during the dark phase of the light cycle, under red light. In this experiment, the focus was on the behaviour of the pup; no attempt was made to record the behaviour of either parent when it was away from the pup. As soon as the observations of that pup were completed, it was replaced in the holding cage (if there were other pups in the litter); after another 2 min, a second pup was introduced into the home cage. No litters were used consisting of more than two pups. If these were opposite sexes, then the order of testing was alternated on each testing day. After the second pup had been tested, the litter was replaced, and the cage was returned to the rack.

Behaviour was recorded from the videoscreen using a one–zero sampling method every 15 sec, over a period of 15 min. This method of recording was chosen in order to record as many behaviour patterns as possible, some of which are of quite brief duration (5–10 sec). It does have the disadvantage that the error can vary if the mean bout length varies—with, say, increasing age of the animals (Martin & Bateson, 1986): this could be a problem for those behaviour patterns that were not brief, such as grooming or nursing. Preliminary observations, however, did not suggest that the bout length varied significantly with age for the measures we used, at least over the age-range studied.

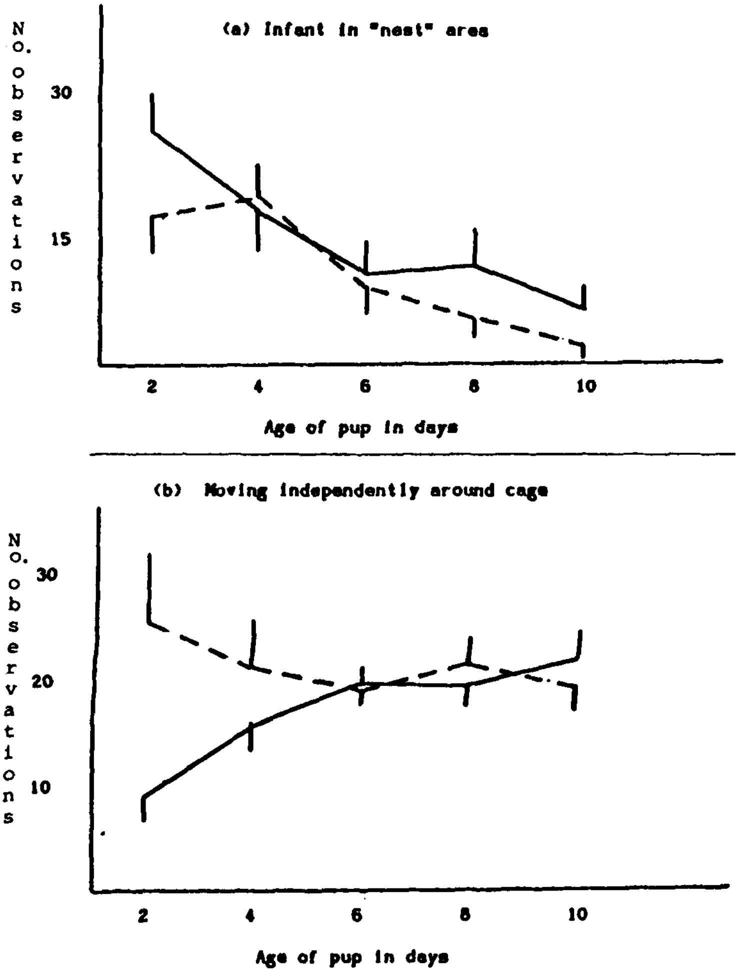
All behaviour of the pup was recorded, as well as behaviour of either parent toward the pup that involved direct contact (licking or carrying). Behaviour patterns that occurred sufficiently often for further analysis were: pup in "nursing area"; pup moving independently around the cage; pup suckling on the mother; pup approaches either mother or father; pup rears up; pup grooms itself; pup feeds independently; pup carried by mother; and pup licked/sniffed by either parent.

Results

Data are summarized in Figure 2 (a–d) and (e–h). The results were analysed using two-way analyses of variance with repeated measures. The following behaviours showed significant age-related changes: feed; $F(4, 120)=4.6$, $p<0.05$; nursing area $F(4, 120)=6.32$, $p<0.01$; carried by mother $F(4, 120)=7.9$, $p<0.001$; rear $F(4, 120)=27.2$, $p<0.001$; and approach father $F(4,120)=4.82$, $p<0.01$. Movement around the cage showed a significant Age × Sex interaction, $F(4, 120)=3.98, p<0.01$, but no significant main effects. Two measures showed both significant age and sex effects: licked by mother, age, $F(4, 120)=2.8$, sex, $F(1, 30)=4.2$, both $p<0.05$; and approach mother, age, $F(4, 120)=12.9, p<0.01$, sex, $F(1, 30)=4.2, p<0.05$. Groom, lick by father, and suckling showed no significant effects on either variable.

Discussion

The results suggest two patterns of behaviour that differentiate the sexes, at least at some ages: female pups are, at all ages, more likely than males to approach the mother, whereas mothers tend to lick males more, particularly when the pups are very young. As might be expected, behaviour associated with parental caregiving (such as licking by a parent, carrying, or nursing) decreased over time, and measures of independent behaviour by the pup (such as movement about the cage, feeding and rearing tended to increase. The rates of increases are not, however, the same for all patterns of behaviour. Measures of independent movement, such as rearing or movement about the cage, increased most between Days 2 and 4, whereas independent feeding increased slightly later, around Days 6–8, as did approaching the mother.



The different responses of the mother towards male and female pups parallels responses observed for rats by Moore and Morelli (1979): mothers seem to direct more licking towards males than females, especially when they are very young. Interestingly, inspection of the raw data for any effect of litter composition suggested that the sex difference may be confined to litters consisting either of singletons or of pups of both sexes. We had, unfortunately, too few data to subject that to statistical test. It does, however, suggest that the responses of *Acomys* mothers to pups are differentiated

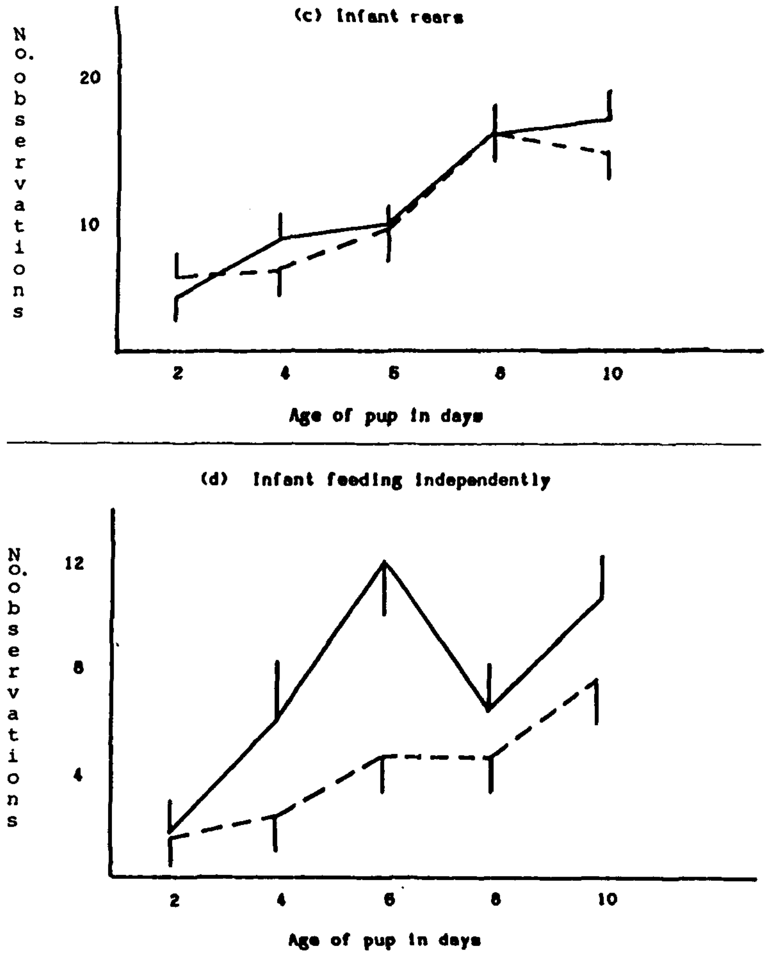
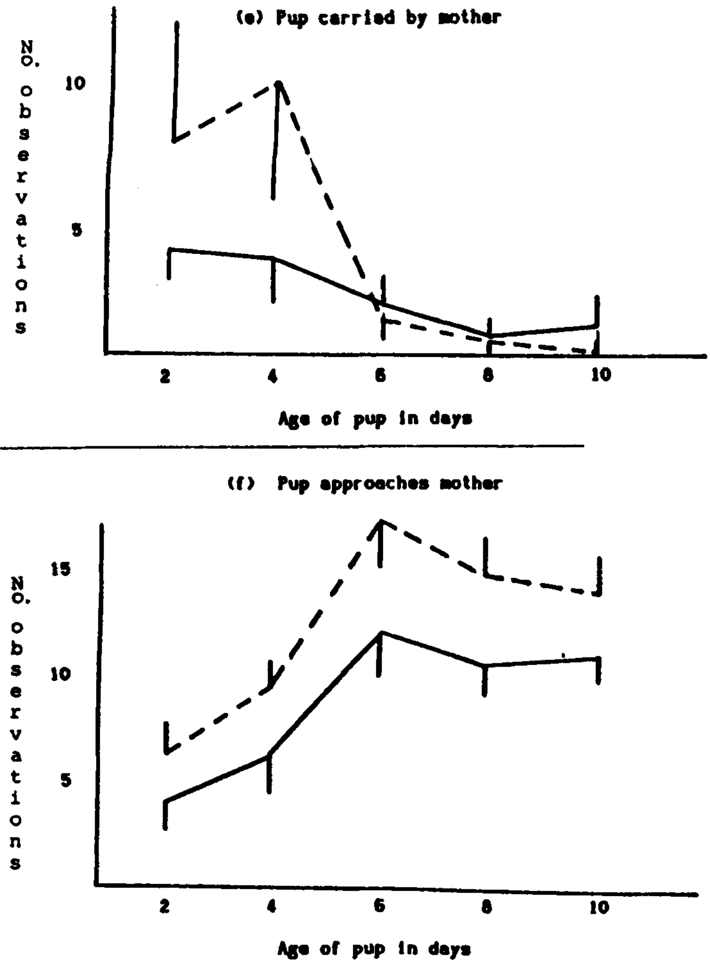


FIG. 2 (a–d). Frequency with which infants engaged in different activities in the home cage: mean frequency ± s.e.m. Solid lines: males ($n = 17$); dotted lines: females ($n = 15$).

within the litter, rather than reflecting absolute differences across all litters—that is, single female pups and females within mixed-sex litters receive relatively less maternal licking than do males in general. Pups in litters consisting of two or more females, on the other hand, seem to receive about the same amount of licking as do males.

These preliminary observations, then, suggest that there are differences between male and female pups in the extent to which they move around the home cage (which may or may not be related to exploration in an unfamiliar



environment). There are also differences in the social experiences of pups of each sex. By contrast, the main experiment focused upon the infant's patterns of exploratory behaviour in an unfamiliar environment in relation to the behaviour of its mother. In this experiment, we restricted social interactions to those between one pup and its mother, partly because interactions with fathers did not show significant effects in the preliminary experiment, and partly for simplicity.

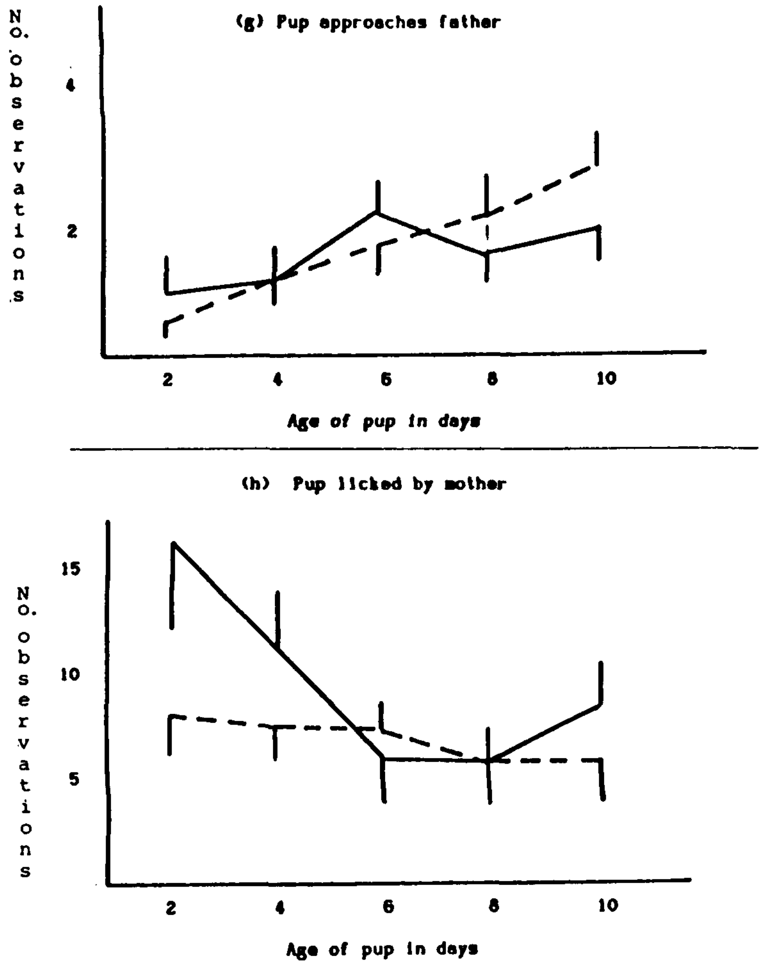


FIG. 2 (e–h). Frequency of social interactions between pup and parents in the home cage: mean frequency ± s.e.m. Solid lines: males ($n = 17$); dotted lines: females ($n = 15$).

EXPERIMENT 3

Methods

Subjects. Twenty-nine infant *Acomys* were used in this experiment, 15 males and 14 females. Six of these were singletons, and the rest were from litters of two or three pups. They were housed and fed as described in the previous experiments; but in this experiment, all cages were supplied with an

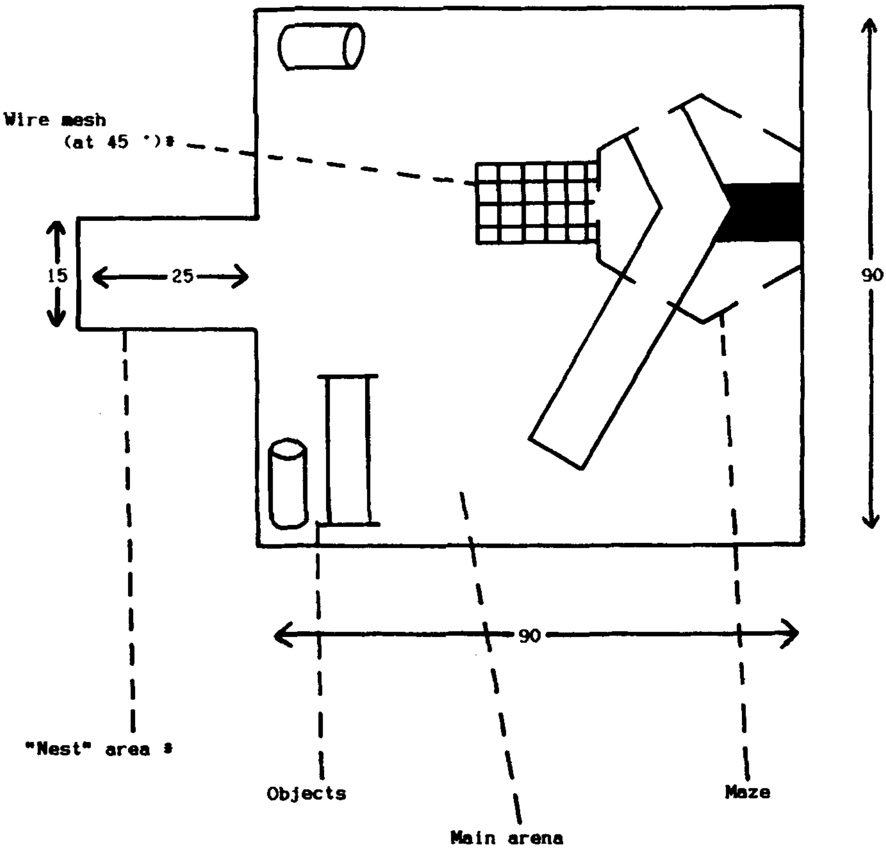


FIG. 3. The "complex environment" used in Experiment 3, to which infants and mothers had access from the "nest" area. The wire mesh was placed at 45° to the floor, so that the animals could climb up it and along the top edge of the maze. The maze was accessible at four doors; the black part was inaccessible during this experiment.

opaque Perspex jar 8 cm long × 5 cm diameter, in which nursing mothers tend to remain with their pups (personal observations).

Apparatus. The testing arena was 90 × 90 × 15 cm and contained various objects for the animals to explore (see Figure 3 for plan). It provided three separate spaces. The "startbox" was 25 × 15 × 15 cm and led directly into the main arena without a door. For testing, animals were placed in here inside their "nest" jar, along with a handful of soiled bedding. The startbox thus comprised a relatively enclosed space, olfactorily somewhat similar to the home cage. This was intended to offer a compromise between "forced" exploration (Welker, 1957) in which the animal is exposed simply to an unfamiliar environment, and "free" exploration, in which it moves from familiar territory. Allowing direct access from home cages, however, would have necessitated disturbing the home cage; we considered this to be undesirable when pups are very young.

In the half of the arena opposite the startbox, there was a small maze adapted from a radial maze around a central hexagon used in previous experiments (see Birke & Sadler, 1987a), which provided the second space. This was divided up so that the animals could enter it in three places, one of which led to a runway of total length 35 cm. This was thus a fairly enclosed space, but one that would not smell familiar and to which access was possible only by traversing the rest of the arena, which comprised the third, relatively open, space.

Along one wall of the maze, facing the startbox was a plastic-coated metal grid placed at 45° to the maze wall, along which the animals could climb to the top edge of the maze. In the other half of the arena were two closed cylindrical objects, 10 and 15 cm in length. The entire arena was placed vertically underneath a videocamera capable of operating at low light intensity.

Procedure. On the day a litter was born, mother and pups, inside the nest jar, were placed in the open part of the arena for 10 min pretrial exposure. No recording was made during this period. Testing began the following day (Day 2) and was repeated on Days 4, 6, 8, 10, and 12.

The behaviour of the animals during the test was recorded on videotape, using a Panasonic videocassette recorder, for a period of 20 min. At the end of that time, the animals were removed from the apparatus and returned to their home cage. The main arena was then washed down with detergent, along with the startbox if animals from another litter were subsequently to be tested. Whenever animals were tested from the same litter, the order of testing animals of one or other sex was varied systematically to control for possible order effects. After a period of 10 min, the mother was returned with the second pup (if any), and filming recommenced.

Behaviour was recorded from the videotaped records, using an Apple II microcomputer. The amount of time spent by each animal in various measures of exploratory behaviour was recorded, as well as time spent in different categories of social interaction. The measures of exploratory behaviour were: time spent in each of the three spaces when the subject animal was not interacting socially (defined operationally as being 15 cm or more from the other animal); sniff objects; time spent climbing metal grid. Measures of social interaction were: following the other animal in each of the three spaces, where following is defined as one animal maintaining a distance of 5 cm or less while the other animal is moving away from it; approaching in each of the three areas, where the other animal is stationary or moving towards; and sniffing/licking the other animal in each of three areas. Time spent by the mother in nursing or carrying the infant was also recorded.

Results

All data were logtransformed for analysis and then analysed using a two-way analysis of variance with repeated measures (for the age of the infant). In order to simplify data from a large number of variables, not all the data are presented. Increases with age in most exploratory behaviours measured would be expected, as would a decline with age in carrying and nursing. There were significant effects due to age on nearly all measures, at a level of $p<0.01$; these are not presented if the change was an expected increase or decrease over all testing days (for example, an increase in time spent in exploring the maze by infants). Table 1 presents a summary showing the direction of age-related effects, rather than presenting data in detail. Data showing significant deviations from the expected are depicted in Figures 4–7. Figure 4 summarizes the exploratory behaviour of the mother, Figure 5 the mother's social behaviour towards the pup, Figure 6 the exploratory behaviour of the pups, and Figure 7 the social behaviour of pups towards the mother.

The results, as summarized in Table 1, were as follows:

1. (a): There were no significant effects of either sex or age of pup on the following measures: *mother*—time apart in arena; sniffing objects; *infants* approach mother in startbox.

2. (b): There were significant sex differences without age effects for: *mothers*—for sniff/lick in maze, males were licked more than females, $F(1,27)=5.4$, $p<0.05$; for time in maze, mothers of males spent more time, $F(1,27)=4.6$, $p<0.05$; and for follow in startbox, males were followed more than females, $F(1,27)=4.2$, $p<0.05$. There were no significant sex differences for *infants* without age effects.

3. (c): Sex differences with age effects were found on the following measures: *mothers*—climbing metal grid, mothers of females climbed more, $F(1,27)=11.4$, $p<0.01$; following in arena, more for mothers of males, $F(1,27)=4.6$, $p<0.05$. *Infants*—follow in startbox, females more than males, $F(1,27)=3.9$, $p<0.05$; sniff/lick mother in arena, females more, $F(1,27)=6.1$, $p<0.05$; sniff object, females more, $F(1,27)=4.0$, $p<0.05$; time in arena, females more, $F(1,27)=4.4$, $p<0.05$.

Discussion

As expected, there were sex differences in exploratory behaviour: females tended to spend longer than males in the open part of the arena, and to make contact with the objects more frequently. Many of the changes in the

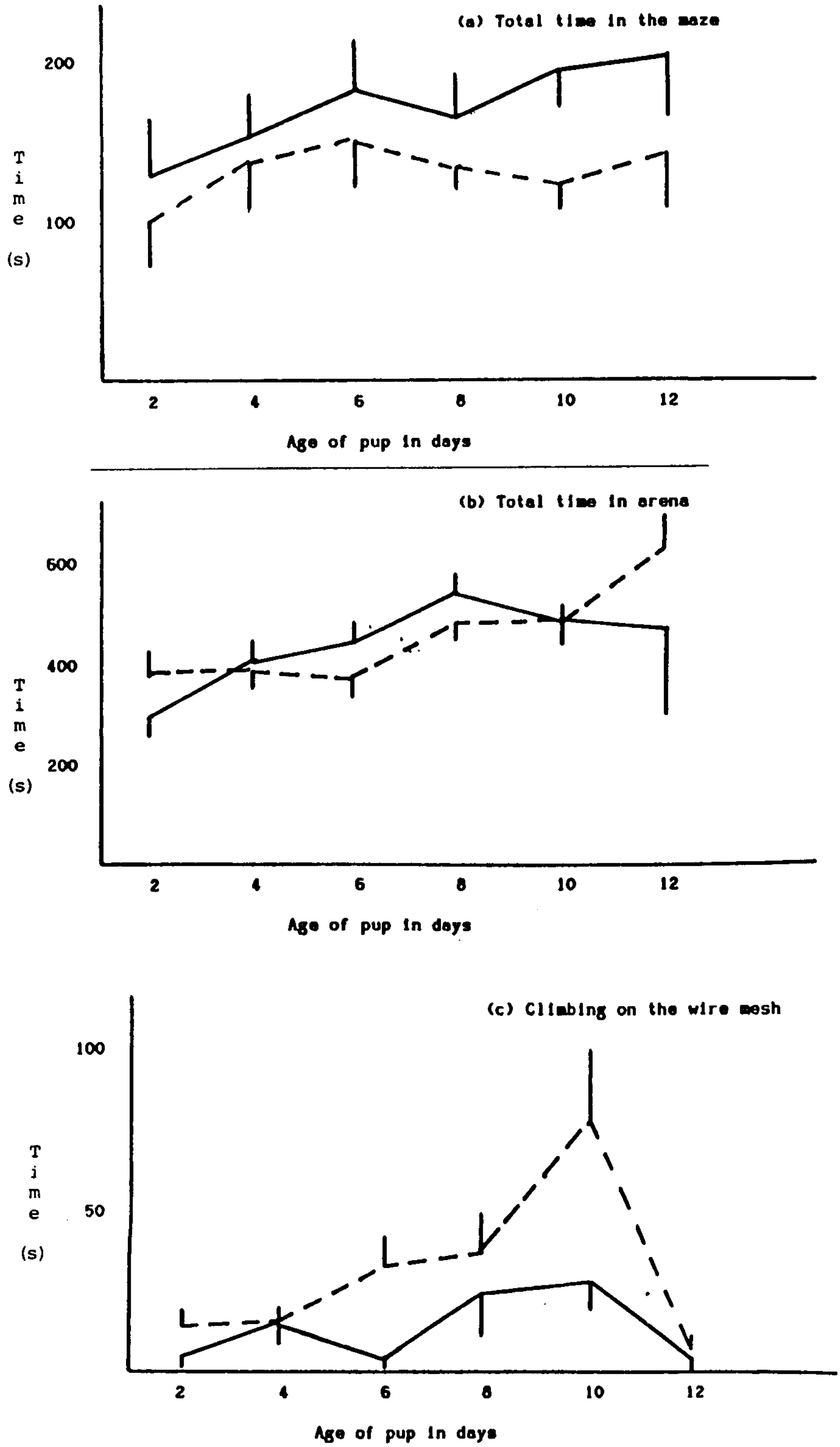
TABLE 1
Summary of Main Effects in Experiment 3

	Behaviour of	
	Mother	*Infant*
(a): Behaviours showing no age or sex effects		
	Time in arena Sniff objects	Approach in startbox/maze
(b): Behaviours showing sex differences alone		
	Sniff in maze (♂>♀) Time in maze (♂>♀) Follow in startbox (♂>♀)	NONE
(c): Behaviours showing age-related changes		
Increased	In maze In startbox alone	In arena (♀>♂) Sniff objects (♀>♂)
Decreased	Approach in startbox Carry infant Sniff infant in startbox	In startbox alone
Increase followed by decrease Age of pup at peak:		
4 days		Sniff mother in startbox
6 days	Approach in arena/maze Follow in arena (♂>♀) Follow in startbox Sniff in maze	Sniff mother in maze Follow in startbox (♀>♂) Sniff in startbox
8 days	Sniff in arena Follow in maze	Sniff mother in arena (♀>♂) Follow in arena
10 days	Time climbing (♀>♂)	Time in maze Sniff mother in maze.

Note This table summarizes the results of Experiment 3 for the behaviour of the mother *(left-hand column) and the infant (right-hand column). (a) shows those behaviours that showed no significant effects; (b) shows those behaviours that showed only differences according to the* sex of the infant; (c) shows those behaviours that changed with age (increase, decrease, or peaking at particular ages). Where these behaviours also showed a difference with sex of pup, this is indicated.

mother's behaviour toward the pup with time are ones that might be expected; pup-carrying by the mother, for instance, decreases with pup age. As the pups get older, mothers tend to spend less time with the pup in the startbox, say, and more time in it alone.

Any tendency for pups to spend more time in the open part of the arena, or to investigate the objects, as they get older may in part be due to increasing


(a) Total time in the maze
200
100
Time (s)
2 4 6 8 10 12
Age of pup in days
(b) Total time in arena
600
400
200
Time (s)
2 4 6 8 10 12
Age of pup in days
(c) Climbing on the wire mesh
100
50
Time (s)
2 4 6 8 10 12
Age of pup in days

familiarity with these stimuli. The pretrial habituation period can only partly compensate for this. Significant order effects due to habituation to the testing situation are, however, less likely to explain differences between males and females or to explain changes in social behaviour.

The data suggest that, on the whole, mothers direct more of their behaviour towards male pups than to females, whereas female pups are both more likely to explore on their own (they are more likely to enter the arena, for instance) and more likely to direct behaviour towards the mother (for example, following, or sniffing mother in the arena). These tendencies are, moreover, most apparent around Days 6–8.

Approaches to pups by mothers were more frequent towards males in all parts of the arena. "Approaches" were defined *a priori* as any movement towards the other animal that resulted in the animals being 5 cm or less apart. Originally, we did not specify the direction of movement in this definition. We did, however, notice from the tapes that some mothers were, in effect, "blocking" the movement of the pup by moving themselves directly in front of the pup as it moved forward, perpendicular to the pup's direction of movement. The pup had then either to stop or to alter direction. A *post hoc* analysis of such "blocking" behaviour showed that 9 of the 14 mothers displayed it, and that it occurred only in the main arena. Interestingly, although "blocking" was relatively infrequent, it occurred more often towards male pups than to females ($p=0.05$; Mann–Whitney Test to compare scores for each sex combined across ages). The behaviour was too infrequent to subject it to more detailed analysis; it is, however, noteworthy that all of the 6 "blocking" mothers that had a pup of each sex in the litter blocked the male pup considerably more often.

In the previous experiment, mothers behaved as rats have been shown to do—that is, they direct significantly more licking to males than to females. In this experiment, one measure of licking/sniffing behaviour did show a significant sex difference; males were licked more than females while in the maze, especially on Day 2. In part, this early differentiation by sex may have been due to mothers reacting differently to males and to females: although at this age male pups were less likely than females to leave the startbox of their own accord, mothers were observed to remove them on several occasions, often carrying them to the maze. So, one reason for high rates of pup-licking in the maze may be that mothers carried very young male pups there.

Carrying behaviour did not show an overall sex difference, but there was a very marked difference at 2 days old: mothers carried female pups for, on

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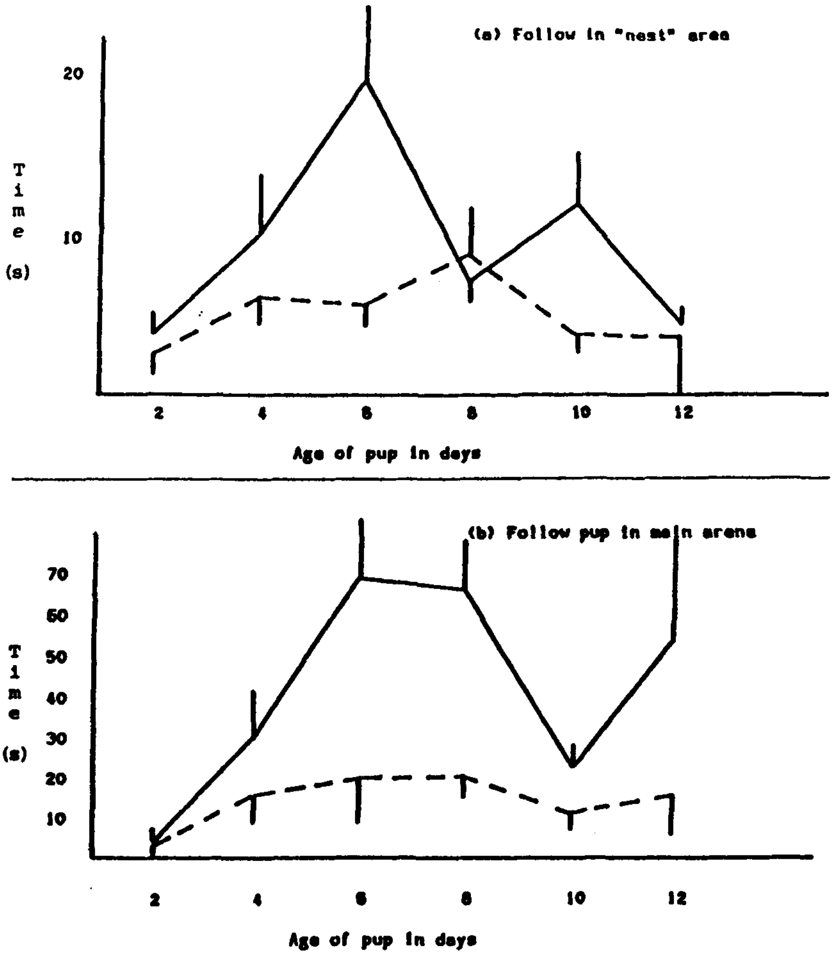
FIG. 4 (a–c). Exploratory behaviour of mothers with pups of different ages (Experiment 3). Solid lines: data for mothers with male pups; dotted lines: data for mothers with female pups. All data are mean times for all mothers ± s.e.m.



very marked difference at 2 days old: mothers carried female pups for, on average, 27 sec, compared to an average of 112 sec for male pups. Scores for nursing behaviour similarly did not show any sex difference, although the data did suggest that there was a substantial difference between males and females at Days 6 and 8. These data were compared between the sexes using *post hoc* t-tests; for Days 6 and 8 only, there were significant differences between the sexes in the amount of time spent nursing (Day 6, males $= 68 \pm 20$ sec, females $= 32 \pm 12$ sec, $t = 2.1$; $p < 0.05$: for Day 8, males $= 55 \pm 12$ sec, females $= 17 \pm 6$ sec; $t = 2.35$; $p < 0.05$), males being nursed more than females.

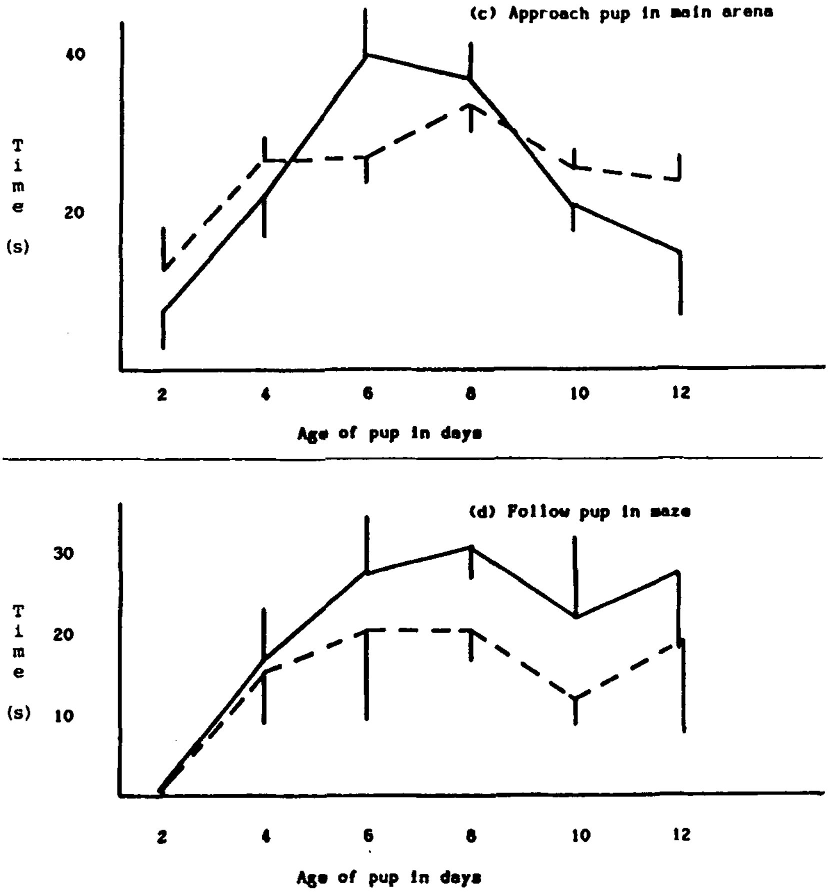
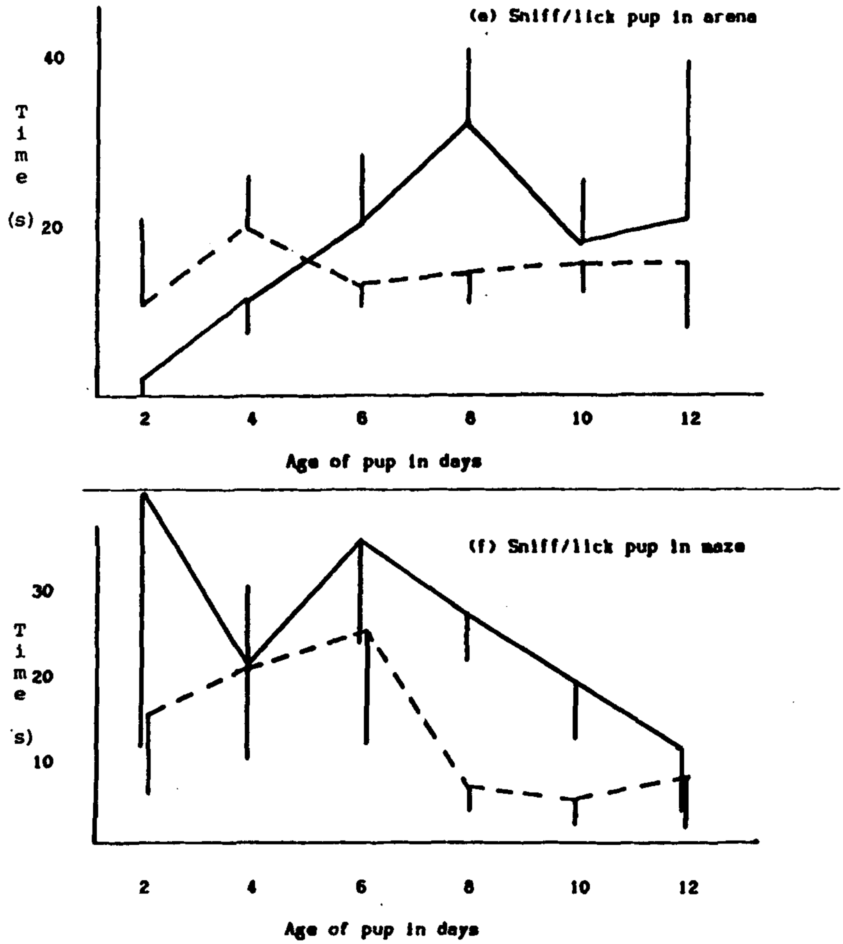


FIG. 5 (a–d). Mean durations of following/approaching behaviour of pups by mothers in Experiment 3. Solid lines: mothers of male pups; dotted lines: mothers of female pups. Mean ± s.e.m.

GENERAL DISCUSSION

Male and female pups appear to differ in their tendency to explore a relatively novel environment. Whether tested alone (Experiment 1) or with the mother present (Experiment 3), females were more likely than males to



explore an open arena. Among very young animals (less than one week old), females moved around more than males, even in the home cage with the parents present. Two-day old males, on the contrary, often ended up in the open arena of Experiment 3 simply because mothers had removed them from the startbox.

Such differences could be related to prenatal hormonal differences between the sexes; little is known, however, about the role of hormones in relation to behaviour in this species. What we can say from the results of this study, on the other hand, is that the observed sex differences in pup

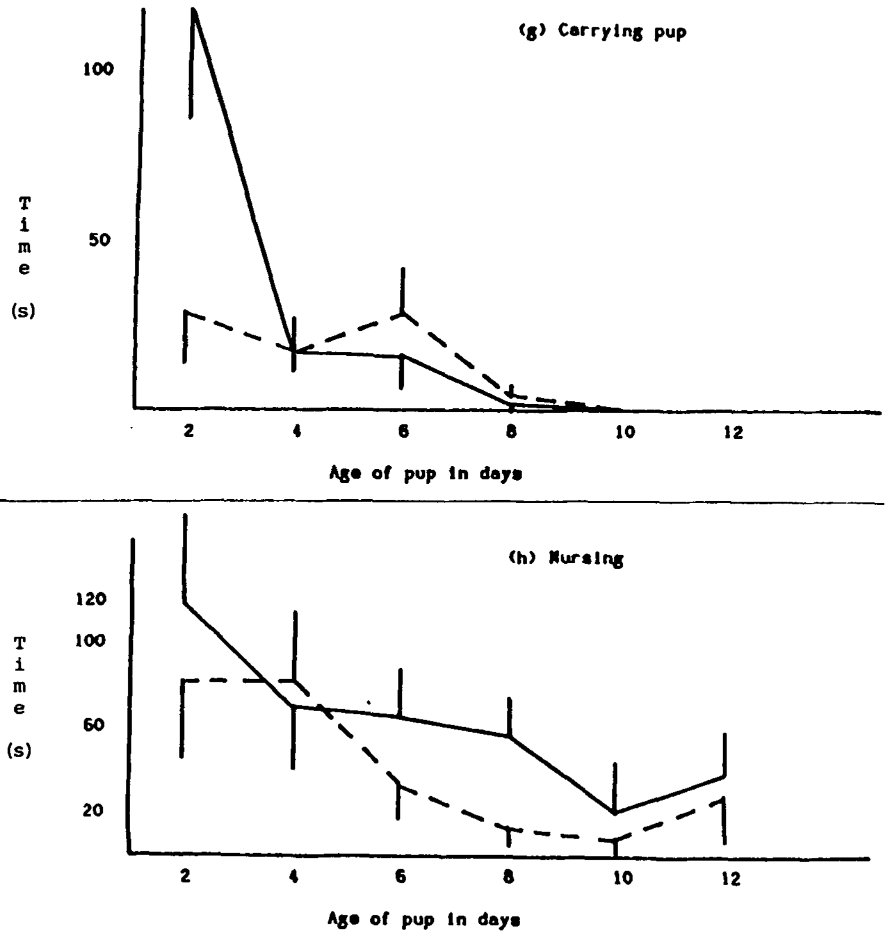


FIG. 5 (e–h). Interaction of mothers with pups of different ages in Experiment 3. Solid lines: mothers of male pups; dotted lines: mothers of female pups. Mean durations ± s.e.m.

behaviour are correlated with discriminatory behaviour by mothers. Mothers generally direct more of their behaviour towards sons than daughters, a tendency that is particularly pronounced around the end of the first week. In particular, mothers make significantly more approaches to sons than to daughters; and, where blocking is shown, it is more likely to be males that are blocked. To what extent such discrimination contributes to—or is perhaps largely responsible for—the emergence of sex differences in pup behaviour raises intriguing questions.

Moore's series of experiments on mother rats (1982, 1984) suggests that

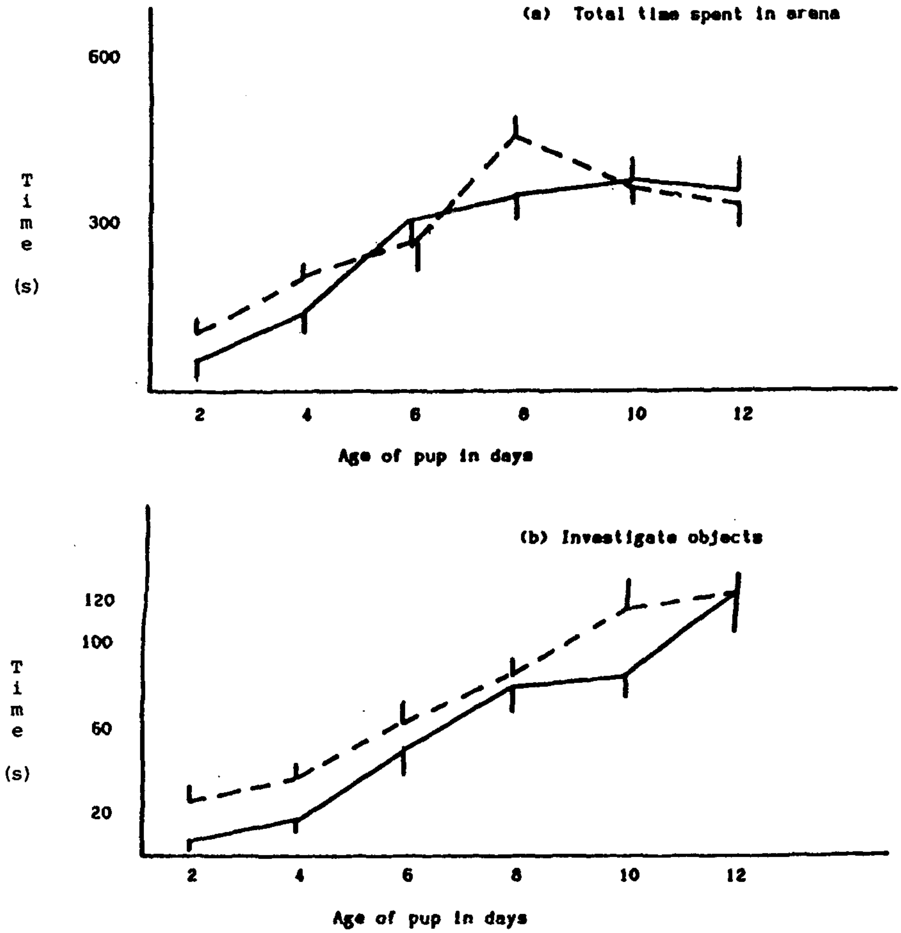


FIG. 6 (a–b). Exploratory behaviour of pups at different ages. Solid lines: males; dotted lines: females. Mean duration (seconds) ± s.e.m.

mothers devote more anogenital licking towards males, and that this is dependent upon odours resulting from metabolites of testosterone in the urine of pups. Experiment 1 showed that, even in the home cage, *Acomys* mothers behaved similarly, directing more anogenital licking toward males.

Yet maternal preference for the odour of males is unlikely to be the sole explanation. Significant differences in the rate of licking by mothers to male and female pups were found only for pups in the maze; yet mothers of sons were much more likely to approach or follow in any part of the arena. These

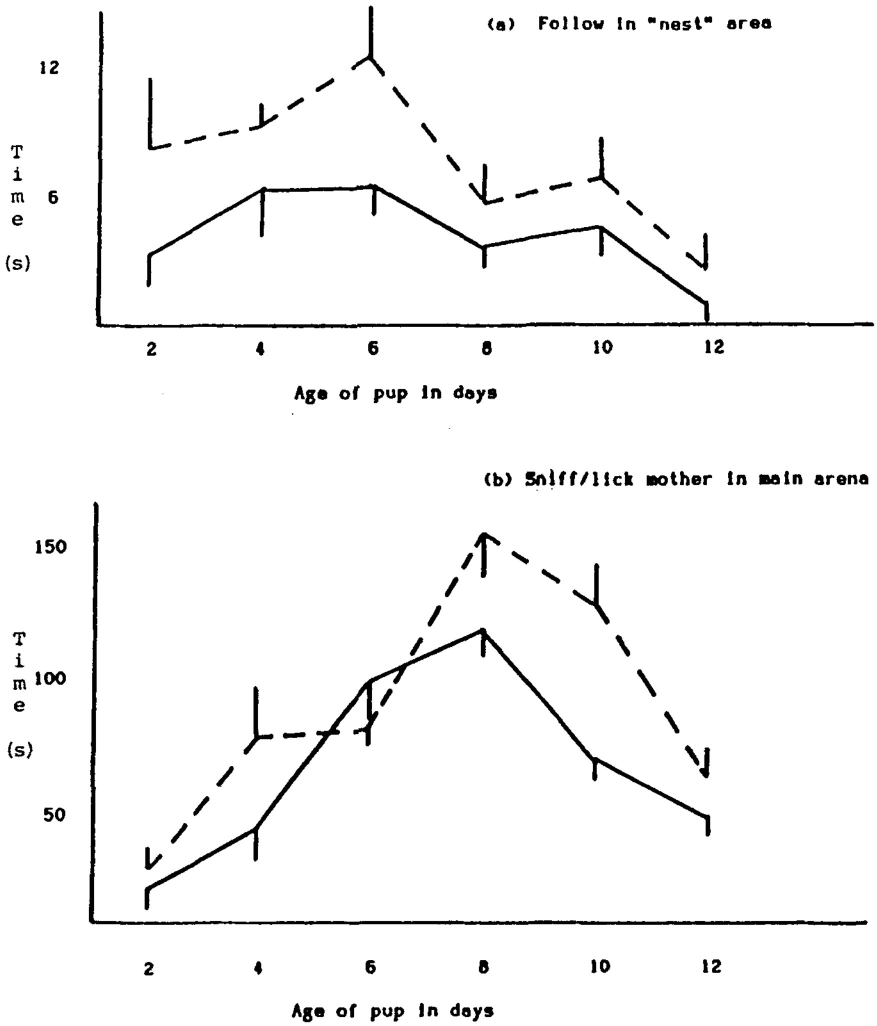


FIG. 7 (a–b). Social behaviour of pups in Experiment 2 towards mothers. Solid lines: males; dotted lines: females. Mean ± s.e.m.

differences, moreover, seem to differentiate most when pups are about one week old.

It is more likely that neither intrinsic differences nor maternal preferences alone can explain the developmental changes. What the data show is a changing pattern of behavioural interaction between mother and pups as the pups get older; but it is a pattern that varies with the sex of the pup. When pups were 2 days old, mothers typically left them in the startbox, inside the "homecage" cylinder, and went off on exploratory forays, returning only briefly to visit the pup. Very young pups usually remained where the mother had put them, so mothers were relatively free to explore. Female pups, however, did sometimes move out into the arena; when they did, mothers typically approached and sniffed at them briefly before continuing on their own forays.

"Blocking" behaviour, though relatively infrequent, is interesting, particularly because it was observed only in the open space of the main arena. Mothers doing this gave the impression that, by intercepting the infant, they were directing it, perhaps to a more enclosed space. Certainly, mothers invariably stopped blocking once the infant entered the startbox or the maze. The tendency for mothers preferentially to block sons raises intriguing questions for the development of exploratory behaviour; females, for instance, might learn to explore more because mothers interfere less with their forays.

Mothers are, of course, only one source of social experience, and under more natural conditions infants interact with many other individuals. It is not yet clear to what extent siblings affect the development of behavioural sex differences in this species. Porter, Cernoch, and Matochik (1983) found that the frequencies of suckling and social contacts by infant *Acomys* were affected by the age of any littermates with whom they were raised. Their study did not report the sex of the pups studied, but it does suggest that the presence of older siblings can be an important influence on early social development.

The present study, while largely descriptive, does suggest that the development of patterns of exploratory behaviour may be partly influenced by how mothers react to infants' exploration. It would be interesting to know more about the long-term consequences, if any, of maternal differentiation in this species. In rats, early maternal differentiation (anogenital licking) has consequences for social and sexual behaviour extending into the juvenile period and even into adulthood (Moore, 1984; Birke & Sadler, 1987b). The present study suggests that, in *Acomys*, mothers differentiate between sons and daughters in various ways over the first two weeks of life: there may well be long-term effects of this discrimination on adult exploratory behaviour.

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La discrimination maternelle et le développement des différences liées au sexe au cours du comportement exploratoire chez la jeune souris épineuse

Cet article a trait aux différences liées au sexe dans le comportement exploratoire chez la jeune souris épineuse, *Acomys cahirinus*. Ces différences pourraient être partiellement causées par des différences dans le comportement maternel envers les jeunes; de même que chez d'autres rongeurs, la mère *Acomys* a un comportement différent envers les jeunes, mâles et femelles. Dans la première expérience, des jeunes *Acomys* exploraient un enclos non familier. Cette expérience montre que même au troisième jour (le premier jour est celui de la naissance), les femelles *Acomys* explorent le nouvel environnement davantage que les mâles. Elles pénètrent dans l'enclos plus rapidement que les mâles et passent plus de temps en contact avec un objet nouveau. La seconde expérience montre que les jeunes femelles sont plus actives que les mâles dans leur cage habituelle, en présence de leurs parents, et s'approchent davantage de leur mère. Les mères, d'autre part, ont davantage de comportements de léchage envers les jeunes mâles. La troisième expérience a trait au comportement exploratoire des jeunes, pris individuellement, en présence de leur mère. Quand elles ont accès à un enclos vaste et complexe, les jeunes femelles explorent davantage que les mâles. Les résultats montrent aussi que les mères ont davantage d'interactions sociales avec leurs jeunes mâles qu'avec leurs jeunes femelles, surtout quand les jeunes sont âgés d'environ une semaine. Certaines femelles paraissent "diriger" les mouvements de leur portée en bloquant leurs déplacements vers l'avant. Cela est plus fréquemment observé envers les mâles qu'envers les femelles. Ces données suggèrent que les changements précédemment observés à ce moment-là dans le comportement exploratoire et l'émergence de différences entre les sexes au niveau de l'exploration pourraient partiellement déprendre des réactions différenciées des mères en fonction du sexe.

Discriminación materna y desarrollo de diferencias sexuales en conducta exploratoria en *Acomys cahirinus*

Este artículo presenta diferencias sexuales en comportamiento de ratones (*Acomys cahirinus*) que pueden estar relacionadas a diferencias en el comportamiento maternal hacia las crías; al igual que otros roedores, las hembras de *Acomys* se comportan diferentamente hacia sus crías de ambos sexos. En el primer experimento, se permitió a infantes de *Acomys* explorar una arena neuva para ellos. Se encontró que ya para el tercer día (respecto al nacimiento) las crías hembra exploraron mas que las crías macho; entraron antes a la arena y pasaron mas tiempo en contacto con un objeto desconocido. El segundo experimento mostró que las crías hembra fueron más activas que las macho cuando se las observó en su jaula habitual en presencia de sus padres, y que iniciaron más acercamientos a la madre. Las madres, por su lado, dirigieron más lamidos hacia los machos. El tercer experimento estudió el comportamiento exploratorio de crías en presencia de sus madres. Cuando se les dió acceso a una arena extensa y compleja, las hembras exploraron más que los machos. Las madres dirigieron más interacciones sociales hacia los hijos que hacia las hijas, especialmente cuando las crías tenían alrededor de una semana de edad. Algunas madres parecieron "dirijir" el movimiento de sus crías bloqueando sus movimientos hacia adelante; estos bloqueos fueron dirijidos más hacia machos que hacia hembras. Los resultados sugieren que los cambios en comportamiento exploratorio observados a esta edad y el surgimiento de diferencias sexuales en exploración pueden depender al menos en parte de las reacciones de las madres hacia sus crías.